

programmable flash memory, 4kB
SRAM, 4kB EEPROM, 8-channel 10-

bit A/D converter and JTAG interface for on-chip debugging. The device

operates between 4.5V and 5.5V.

Ultrasonic distance sensor. This

nic distance sensor. This sensor is a modified version of the original UDM sensor. The UDM sensor module has four pins but this module has only three pins for connection to external circuit. The transmitter unit sends an ultrasonic wave of 40kHz directed towards the target. The reflected signal is received by the receiver unit, which calculates distance based on the speed of ultrasonic waves and the time required by the waves to travel to-and-fro.

It forms a 9-byte frame as xxx.xxxx < new line > of the calculated distance and sends it through a serial (UART) interface to the microcontroller at pin PEO configured as RXD0. The sensor works on 5V DC power supply. Its minimum and maximum ranges are 10cm and 400cm, respectively. The rate of transmission of the serial data is 9600bps with TTL level output. The ultrasonic distance sensor is shown in Fig. 3.

The sensor can be easily interfaced with any microcontroller. Output from the sensor is in a frame of nine bytes. The frame format xxx.xxcm < new line > is described in Table I.

Slotted opto isolator module MOC7811.

Internally, the module consists of an LED and a photo transistor. The plastic package encloses them in such a way that there is gap in between these two components

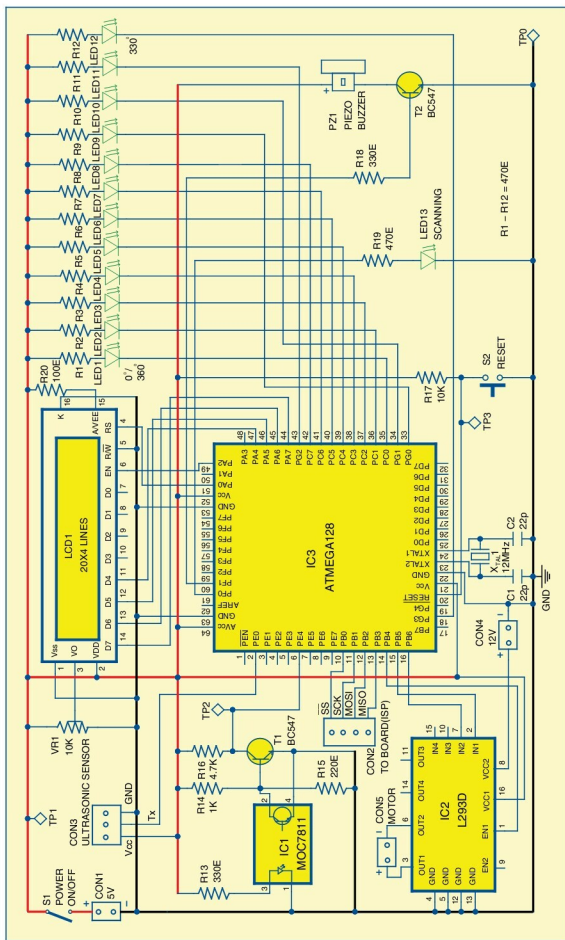


Fig. 2: Schematic diagram of the ultrasonic radar model using microcontroller ATmega128